MIL-I-22668A(AS)
Amendment 4
28 September 1967
Superseding
Amendment 3
15 August 1966

MILITARY SPECIFICATION

Integrated Electronic Centrals AN/ASQ-57, AN/ASQ-57A AN/ASQ-57B, AN/ASQ-109, AN/ASQ-58, AN/ASQ-58A, AN/ASQ-58B and AN/ASQ-110

This amendment forms a part of Military Specification MIL-I-22668A(WEP) dated 15 March 1963 and has been approved by the Naval Air Systems Command, Department of the Navy.

*By this amendment basic MIL-I-22668A(WEP) is changed to MIL-I-22668A(AS) and throughout the specification, Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

*Page 1, Specification Title: Delete and substitute the title of this amendment.

Delete any reference to "AN/ASQ-56(*)" that appears in this specification.

*Page 1, paragraph 1.2: Delete and substitute

"1.2 Classification - The Integrated Electronic Centrals shall be of eight types which together shall include the following equipments:

Item	Name	Type Designation	MTBF	MIL-STD-781 Test Plan
1	Receiver-Transmitter, Radio	RT-542/ASQ	170	III
2	Receiver-Transmitter, Radio	RT-5242/ASQ	170	III
3	Receiver-Transmitter, Radio	RT-541/ASQ	150	III
4	Decoder, Pulse	KY-309/ASQ		
5	Coder-Receiver-Transmitter	KY-533A/ASQ	450	IV
6	Amplifier-Power Supply-Receiver	AM-2310/ASQ	400	IV
7	Antenna	AS-909/ARA-48	2500	VI
8	Control, Radio Set	C-3146/ASQ	6000	V

FSC 5895

Source: https://assist.dla.mil -- Downloaded: 202640-19T13:17Z
Check the source to verify that this is the current version before use.

Item	Name	Type Designation	MTBF	MIL-STD-781 Test Plan
9	Control, Radio Set	C-2790/ARC		
10	Switching Unit	SA-629/ARC	560	V
11	Indicator, Frequency Channel	ID-758/ARC		
12	Mounting	MT-2141/ARC	--	--
13	Mounting-Plenum	MT-2244/ASQ	--	--
14	Control, Radio Set	C-2791/ARC	--	--
15	Control, Radio Set	C-2791A/ARC	--	--
16	Receiver-Transmitter, Radio	RT-559/ASQ-58	150	III
17	Receiver-Transmitter, Radio	RT-559 A /ASQ-58	150	III
18	Control, Radio Set	C-1607/ARC-52	--	--
19	Control, Transponder Set	C-6280(P)/APX	--	--
20	Control, Receiver	C-1457/ARR-40	--	--
21	Mounting-Plenum	MT-3659/ASQ	--	--
22	Coder-Receiver-Transmitter	KY-533/ASQ	450	IV
23	Control, Radar Set	C-1159/APX-6B	--	--
24	Control, Radar Set	C-1272A/APA-89	--	--
25	Control, Radio Set	C-8057/ARC	--	--
26	Equipment	Juliet-28	--	--

*Page 2, paragraph 1.3: Delete table and substitute

"*Items Required for Each Central"

AN/ASQ-57A	AN/ASQ-57B	AN/ASQ-58A	AN/ASQ-58B	AN/ASQ-57	AN/ASQ-109	AN/ASQ-58	AN/ASQ-110
Item 1	Item 2	Item 1	Item 2	Item 1	Item 2	Item 1	Item 2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	6	6	6	6
6	6	6	6	7	7	7	7
7	7	7	7	8	8	8	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:17Z
Check the source to verify that this is the current version before use.

AN/ASQ-57A	AN/ASQ-57B	AN/ASQ-58A	AN/ASQ-58B	AN/ASQ-57	AN/ASQ-109	AN/ASQ-58	AN/ASQ-110
Item 8	Item 8	Item 8	Item 8	Item 9	Item 9	Item 13	Item 13
9	9	14	15	10	10	14	15
10	10	16	17	11	11	16	17
11	11	18	18	12	12	18	18
12	12	19	19	13	13	22	22
19	19	21	21	20	20	23	23
20	20		25	22	22	24	24
21	21		26	23	23		25
	25			24	24		26
	26				25		
					26		

* Item numbers correspond to items of paragraph 1.2.

The C-2791/ARC, C(*)/ARC, C-1607/ARC-52, C-6280(P)/APX, C-1159/APX-6B, C-1272A/APA-89, C-1457/ARR-40, C-8057/ARC and Juliet-28 are part of the Centrals but normally procured separately.

*Page 4, paragraph 2.1, under SPECIFICATIONS Military: Add

"MIL-C-38257 Control, Transponder Set C-6280(P)/APX"
MIL-C-81470(AS) Control, Radio Set C-8057 /ARC and delete

"MIL-M-22673 Mounting Plenum MT-2243/ASQ-56"

Page 4, paragraph 2.1, under SPECIFICATIONS Military, after "MIL-C-22672":

Delete "KY-308(*)" and substitute

"KY-533/ASQ and KY-533A/ASQ"

Make this change wherever "KY-308(*)/ASQ" appears in this specification

Page 4, paragraph 2.1, under SPECIFICATIONS Military: Delete

"MIL-R-23094 Reliability Assurance for Production Acceptance of Avionic Equipment"

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:17Z
Check the source to verify that this is the current version before use.

and under STANDARDS Military: Add

"MIL-STD-781 Reliability Tests, Exponential Distribution"

Page 5, paragraph 3.1.2: Delete

*Page 5, paragraph 3.2.2: Delete and substitute

"3.2.2 Total Weights - The total weight of each central, excluding cables, shall not exceed:

AN/ASQ-57,	250 lbs.	AN/ASQ-58	284.4 lbs.
AN/ASQ-109,	250.5 lbs.	AN/ASQ-110	285.4 lbs.
AN/ASQ-57A,	252.5 lbs.	AN/ASQ-58A	286.9 lbs.
AN/ASQ-57B,	253 lbs.	AN/ASQ-58B	287.9 lbs."

*Page 6, paragraph 3.2.4.4: Delete and substitute

"The Centrals shall have a specified mean (operating) time-between-failure when tested as outlined under the requirements of 4.4.3 of 55 hours for the AN/ASQ-57, AN/ASQ-109, AN/ASQ-57A and AN/ASQ-57B, and 50 hours for the AN/ASQ-58, AN/ASQ-110, AN/ASQ-58A and AN/ASQ-58B. For individual equipments when purchased separately the requirement shall be as shown in 1.2."

Page 7, paragraph 3.2.8: Delete and substitute

"Unless otherwise specified in the individual equipment specification, the generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181B except that an additional 54 db, 43 db and 58 db shall be added to the limits of curves 1, 16 and 17 respectively using a 1 KC bandwidth for tests conducted in accordance with Paragraph 3.1.1 (MIL-I-6181B).

Page 7, paragraph 3.2.12: Delete "30 $\pm$ 10°C" and substitute "25 $\pm$ 5°C".

*Page 8, paragraph 3.2.13: Delete

"Mounting Plenum	MT-2243/ASQ-56	Class 2"

and add

"Mounting Plenum	MT-3659/ASQ	Class 2
Receiver-Transmitter Radio	RT-559/ASQ-58	Class 2
Receiver-Transmitter Radio	RT-559/ASQ-58	Class 2
Receiver-Transmitter Radio	RT-542A/ASQ	Class 2
Control, Radio Set	C-2791A/ARC	Class 1"

Source: https://assist.dla.mil -- Downloaded: 20260-19T13:17Z
Check the source to verify that this is the current version before use.

Page 8, paragraph 3.2.13.1: Delete and substitute

"3.2.13.1 System Cooling Air Requirement - Cooling air requirements for the centrals are given in Figures 1 and 2."

*Pages 8 and 9, paragraph 3.2.15: Make the following changes

After "AN/ASQ-57" add "AN/ASQ-57A, AN/ASQ-109 and AN/ASQ-57B"

After "AN/ASQ-58" add "AN/ASQ-58A, AN/ASQ-110 and AN/ASQ-58B."

Pages 10, 11 and 12: Delete and substitute Figures 1 and 2 of this amendment.

*Page 13, paragraph 3.3.1: After "RT-542/ASQ" add "and RT-542A/ASQ."

*Page 13, paragraph 3.3.2: After "RT-559/ASQ-58" add "and RT-559A/ASQ-58."

*Page 13, paragraph 3.3.7: After C-2791/ARC" add "and C -2791A/ARC."

*Pages 13 and 14, paragraphs 3.3.10, 3.3.11 and 3.3.12: Delete and substitute

"3.3.10 IFF/Coder Control - The remote control function required for the operation of the IFF shall be performed by Control, Radar Set C-6280(P)/APX in accordance with Specification MIL-C-38257, or Control C-1159/APX-6B in accordance with Specification MIL-R-7893 and Control C-1272A/APA-89 in accordance with Spcification MIL-C-7953.

"3.3.11 Radio Set Control - The remote control function required for the operation of the Juliet equipment shall meet the requirements of Control Radio Set C-8057/ARC Specification MIL-C-81470(AS).

*Page 14, paragraph 3.3.13: Add

"The mounting plenum function for the Integrated Electronic Centrals shall meet the requirements of Mounting Plenum MT-2244/ASQ or MT-3659/ASQ of Specification MIL-M-22674."

Page 15, paragraph 4.2.1.1, last sentence: Delete

Page 16, paragraph 4.2.3: Add

"During temperature/altitude tests outlined in Specification MIL-T-5422, (steps 2, 3, 4, 6, 7, 9, 10, 11, 12 and 14) cooling air may be supplied at all times."

Page 16, paragraph 4.2.4, last sentence: Delete all after

"under the contract."

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:17Z
Check the source to verify that this is the current version before use.

Page 18, paragraph 4.4.1.3: Delete sentence 7 and substitute

"The direction of vibration shall be either (a) vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes or (b) rotary for 10 minutes".

Page 19, paragraph 4.4.2: Delete and substitute

"4.4.2 Sampling Tests - Centrals selected for sampling tests shall first have passed the individual tests. Centrals shall be selected for sampling tests by the government inspector in accordance with the following" (Sampling tests shall not be conducted unless Reliability Assurance tests of 4.4.3 are deleted by contractual action.)

Quantity of Centrals Offered for Acceptance	Quantity of Centrals Selected for Sampling Tests
First 10	1
Next 50	One for each additional 100 or fraction thereof"

Page 19, paragraph 4.4.2.1: Delete "equipment and"

Page 19, paragraph 4.4.2.1(2): Delete "equipment or"

*Page 20, paragraphs 4.4.3, 4.4.3.1 and 4.4.3.2: Delete and substitute

"4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. A Qualification Phase and Sampling Phase Tests shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase, Test Level E shall be used. The Accept-Reject criteria for Test Plan III shall be used to determine the length of the tests (until an accept or reject decision is reached).

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:17Z
Check the source to verify that this is the current version before use.

4.4.3.2.1 Lot Size for Sampling Phase - The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be evaluated as one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples of each months production shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime other equipments constructed shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment quality control, etc. so that the entire contract quantity will show an accept decision."

Page 20, paragraph 4.4.3.4(1): Delete "RT-542 and RT-559 and substitute "RT-542, RT-542A, RT-559 and RT-559A" and revise the specification requirements and acceptance requirements as follows:

Characteristic	Specification Requirement	Acceptance Requirement
"Audio Capability	Standard not less than 250 mw	Not less than 62.5 mw
	Service not less than 62.5 mw	
Modulation Capability (input to clipping)	Not less than 80%	Not less than 70%
Tone Modulation Capability	Not less than 70%	Not less than 70%
Side Tone Level	Standard not less than 250 mw	
	Service not less than 75 mw	Not less than 75 mw"

Page 21, paragraph 4.4.3.4(2) - Revise the specification requirements and acceptance requirements as follows:

Characteristic	Specification Requirement	Acceptance Requirement
"Audio Capability	Standard not less than 250 mw	
	Service not less than 62.5 mw	Not less than 62.5 mw"

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:17Z
Check the source to verify that this is the current version before use.

Delete:

"ADF 100 cps Inverter and ADF Motor Control"

Page 21, paragraph 4.4.3.4(3): Delete and substitute

"(3) TACAN-KY-309 and RT-541

Characteristic	Specification Requirement	Acceptance Requirement (if lower)
Distance Accuracy and Sensitivity	± (0.2 mi + 0.1%) for sig. levels from -117 to -30 dbw	± (0.3 mi + 0.1%) for sig. levels from -114 to -30 dbw (20% 15 cycles, 20% 135 cycles
Bearing Accuracy and Sensitivity	±2° with sig. levels from -113 to -30 dbw	± 2.5° with sig. levels from -110 to -30 dbw (20% 15 cycles, 20% 135 cycles).
Identification	Standard not less than 50 mw audio with 30 db S/N ratio for sig. levels from -114 to -30 dbw. Service not less than 50 rw audio with 30 db S/N ratio for sig. levels from -111 to -30 dbw	Not less than 50 mw audio with 30 db S/N ratio for sig. level from -111 to -30 dbw.
Power Output	Standard not less than 30 dbw Service not less than 27 dbw"	Not less than 27 dbw

Page 22, paragraph 4.4.3.4(4): Delete and substitute

"IFF/Coder performance characteristics to be measured on the KY-533/ASQ and KY-533A/ASQ shall be in accordance with specification MIL-C-22672."

Page 23, paragraph 4.4.3.6: Delete and substitute

"4.4.3.6 Determination of Compliance - Compliance shall be determined in accordance with Test Plan III of MIL-STD-781."

Source: https://assist.dla.mil -- Downloaded: 20260-19T13:17Z
Check the source to verify that this is the current version before use.

Page 25, paragraph 4.5.5: Delete "an equipment or" and substitute "A".

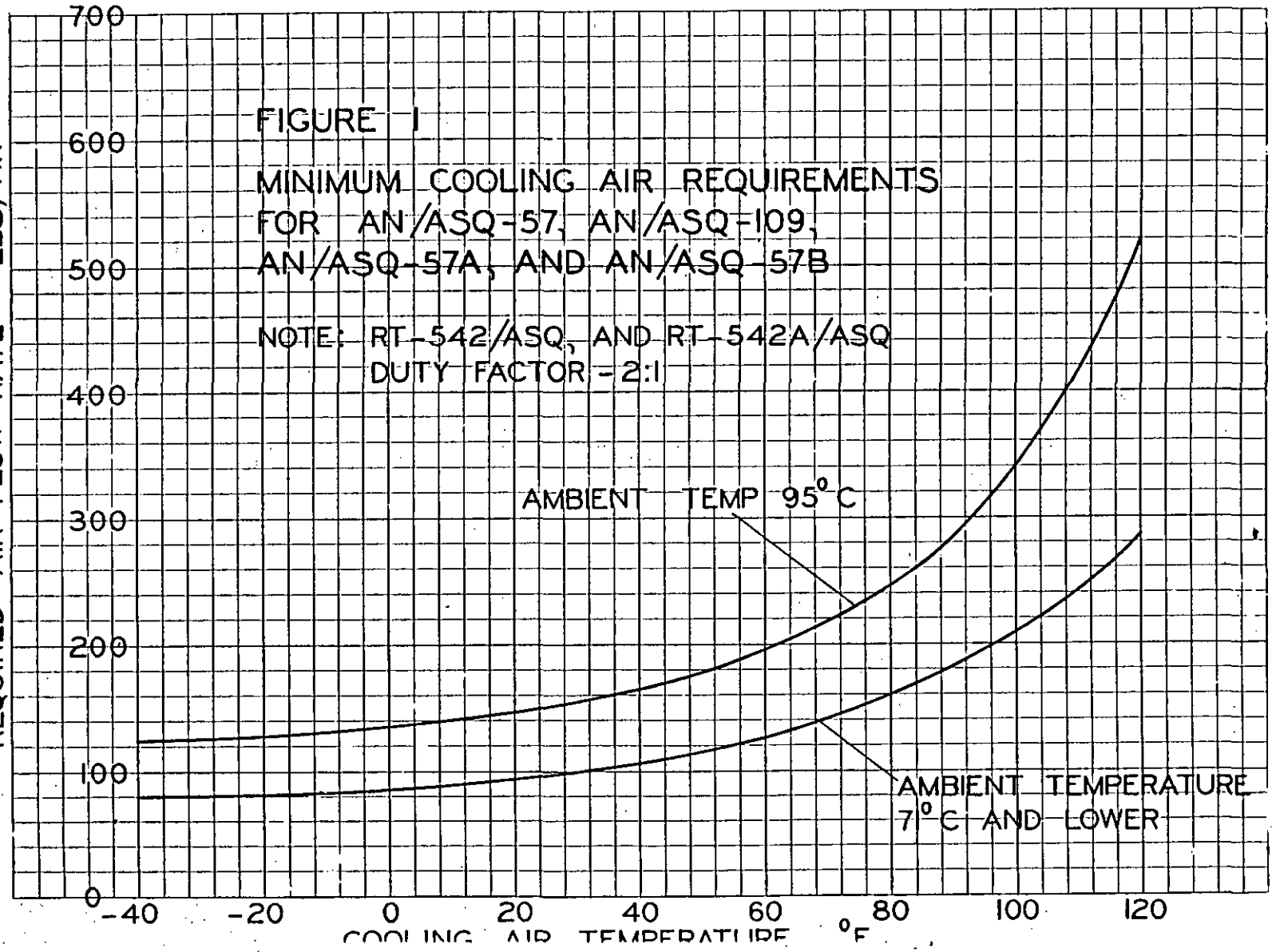
Page 26, paragraph 6.2: Add

"The d-c power requirement shall be as specified in 3.2.1.5 and MIL-STD-704 except that the lower voltage limit under emergency condition shall be 21 volts in lieu of 17 volts."

*An asterisk preceding a page number denotes a change to the specification which did not appear in the previous amendment.

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:17Z
Check the source to verify that this is the currentversion before use.

FREDERICK POST COMPANY
GRAPH CROSS SECTION 8X8 PER INCH



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:17Z
Check the source to verify that this is the current version before use.

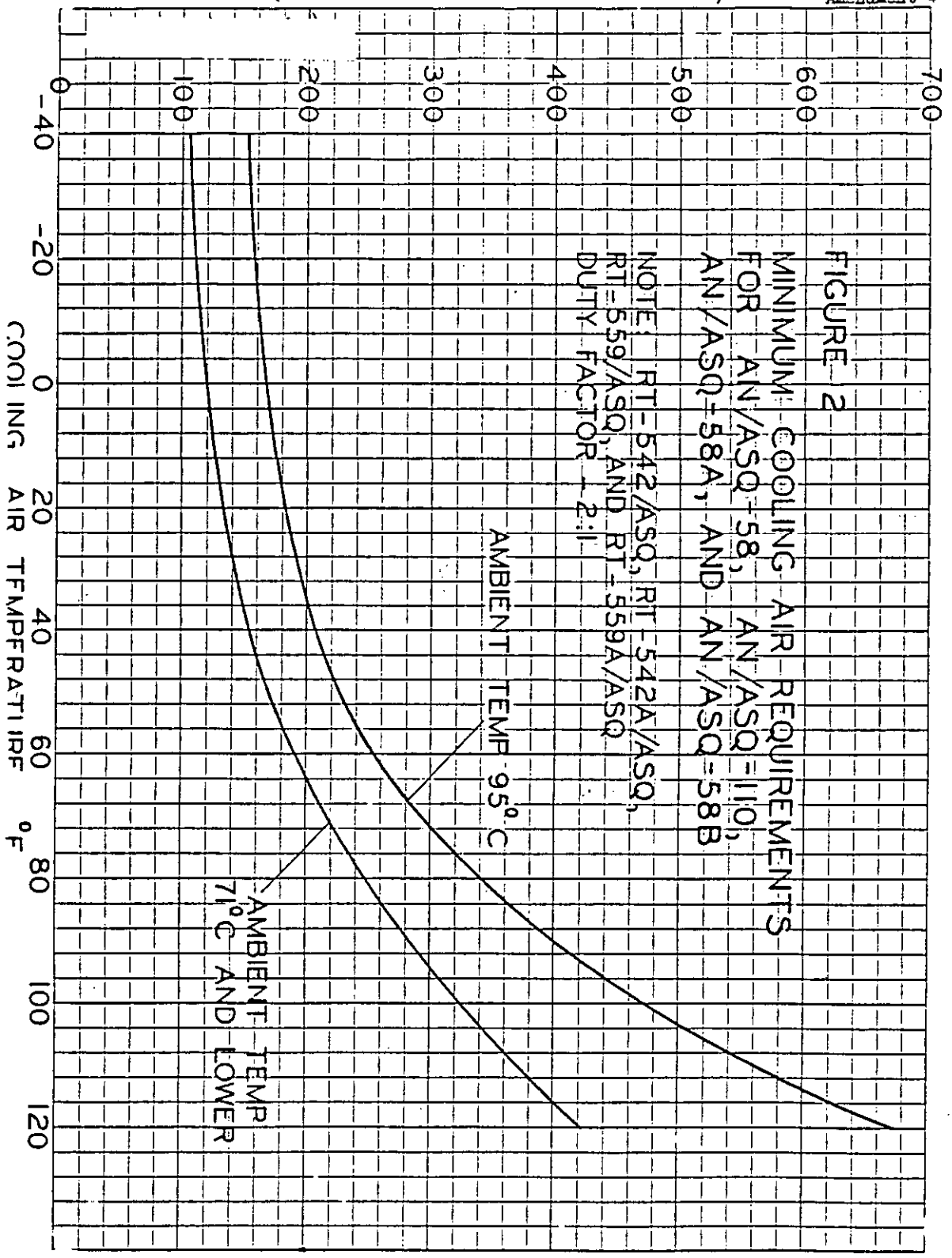


FIGURE 2

MINIMUM COOLING AIR REQUIREMENTS FOR AN/ASQ-58, AN/ASQ-110, AN/ASQ-58A, AND AN/ASQ-58B

NOTE: RT-542/ASQ, RT-542A/ASQ, RT-559/ASQ, AND RT-559A/ASQ DUTY FACTOR - 2:1

FREDERICK POST COMPANY
EDITED CROSS SECTION 880 PER INCH

Source: https://assist.dla.mil -- Downloaded: 20261010-19T13:17Z
Check the source to verify that this is the current version before use.